

Ties that Bind: A Network Approach to Assessing Knowledge Transfers from the ICT Industry

Abstract

We create an inter-sectoral citation network for 1.5 million patents granted between 1976 and 2010 across all industries in the U.S. In this network, we find a significant increase over time in the innovation centrality of the ICT industry as well as in the “closeness” of non-ICT industries to the ICT industry. Such “closeness” of a non-ICT industry to the ICT industry in the innovation network space yields a significant increase in the proportion of ICT technologies, business model patents, and complementary innovations of the non-ICT industry. In turn, closeness results in superior R&D performance of the citing industry. A unit standard deviation increase in the closeness of an industry to the ICT industry yields a 10.2% increase in innovative efficiency, 48.39% increase in recombinant reuse, and 9.09% increase in recombinant creation capabilities. These results point to a paradigm shift in knowledge production, and also have important implications for the nature of competition in non-ICT industries.

1. Introduction

In his letter to shareholders in 2013¹, General Electric (GE) CEO Jeffrey R. Immelt remarked, “*We believe that every industrial company will become a software company.*” As proof points for his vision, he included snapshots of technology-enabled product innovations across different verticals that aimed to improve asset utilization and revenue for GE’s customers. A rich body of anecdotal evidence from industry suggests that GE is not alone in its experience of growing dependence on information and communication technologies (ICT) for product differentiation and performance. Indeed, ICT is embodied in smart screens of a variety of devices ranging from phones to home appliances to medical devices, an estimated 80 percent of car innovations from entertainment to crash-avoidance systems (Charette 2009), and wearable devices that monitor the health and performance of athletes and patients alike. These instances of innovation emphasize that technological progress resulting from R&D performed within the ICT industry generates not only private returns to the ICT industry but also yields social returns from the embodiment of ICT in innovations in other industries. However, there is little empirical documentation of the innovation centrality of the ICT industry and its implications, including shifts in innovation content and outcomes as well as competitive dynamics in ICT-centric industries.

Prior research has documented significant changes to R&D resulting from ICT investments. Indeed, ICTs enable better management of knowledge assets (Thomke 2006), yield greater precision and reduction of costs in the production of innovations through better tools for product test, design and simulation (Nightingale 2000), and foster inter-organizational coordination and knowledge acquisition (Von Hippel and Von Krogh 2006). These changes to innovation processes yield significant improvements in the speed and efficiency of innovation. Indeed, Kleis et al. (2012)) find for a given level of innovation spend, a 10% increase in IT capital relates to a 1.7% increase in the number of patents. However, they also note that the coefficient for the interaction between IT and R&D is insignificant. In turn, they conclude that while the IT capital of a firm increases its quantum of patents, it does not increase R&D productivity. More recently, Branstetter et al. (2019)) expand the focus of this

¹ "Letter to Shareowners." *GE 2013 Annual Report*; https://www.annualreports.com/HostedData/AnnualReportArchive/g/NYSE_GE_2013.pdf (last accessed on Nov 09, 2020).

literature beyond IT capital to examine more specifically the incorporation of ICT in R&D by measuring the digitization of products and product development. They find a software-biased shift in innovation, but their evidence rests on an increase in citation intensity of software patents alone in four manufacturing industries. Further, they find that such shifts in innovation increase R&D productivity and firm value but offer limited insights into the underlying mechanisms. Our study advances this rich body of work on ICT and R&D by examining the contribution of the ICT industry that includes not just software but also, networks, devices and other electronic equipment to innovation in not just manufacturing industries but all non-ICT sectors of the economy. Our study additionally documents modal shifts in knowledge processes and outcomes and competitive dynamics in the using industries.

Using data on 1.31 Million US patents granted between 1981 and 2010, we construct the network of all patent citations across all sectors of the economy. We then estimate for each year, the centrality of each industry in the inter-sectoral citation network. Subsequently, we investigate three main questions: 1) For each year in the sample time period, what are the industries with the highest centrality in the innovation network? 2) What technological shifts and R&D outcomes are enabled in the citing industry by the central industries? 3) What are the implications for competitive dynamics?

Our analyses demonstrate that the embodiment of the ICT industry's technologies in a range of disaggregated sectors is significant, systematic, and pervasive. Specifically, we find a remarkable increase over time in the centrality of the ICT industry in the inter-sectoral citation network, as manifest in increased citations from diverse industries to technologies owned by the ICT industry. The "closeness" of a non-ICT industry to the ICT industry in the innovation network space is associated with a significant increase in the proportion of ICT technologies, complementarity in technologies, and business model innovations of the non-ICT industry. In turn, closeness to the ICT industry results in greater R&D productivity or patents per R&D dollar, and recombinant creation and reuse. We also provide preliminary evidence of a shift in the competitive dynamics of industries. In particular, we find that industries that are closer to the ICT industry in the innovation network demonstrate greater volatility and performance spread than peripheral industries.

Our analyses correct for the endogenous nature of the citation network through an Instrumented Network approach, where we estimate the network using Exponential Random Graph Models (ERGMs)

with patent examiner characteristics as an exogenous source of variation in patent citations. We also demonstrate the robustness of our results to alternate variable measures, empirical specifications, and exogenous sub-networks in the data. Our study documents and demonstrates the dominant role played by the ICT industry in innovation across all industries of the economy. The network structure of innovation informs our understanding of key technological opportunities offered by the ICT industry. This structure also provides insight on how these opportunities manifest in the central and peripheral non-ICT industries. In turn, our results emphasize fundamental shifts in knowledge production and competitive dynamics of various industries.

The rest of this study is organized as follows. Section 2 presents testable predictions about a technology-biased shift in R&D and the implications of such a shift. Section 3 describes our data and Section 4 describes key variables and network measures. Section 5 articulates our empirical strategy, notably, the use of ERGM and instrumented network, to correct for endogeneity of network formation. Section 6 discusses the main results. Section 7 presents robustness checks and Section 8 concludes.

2. Theory

2.1 The General-Purpose Nature of ICT

A rich body of work points to the criticality of ICTs in innovation and knowledge creation in the firm (Quinn et al. 1997, Kleis et al. 2012). These studies document several evolutionary changes to R&D engendered through ICT - better management of knowledge assets (Thomke 2006), greater precision and reduction of costs of producing innovation through better tools for product test, design and simulation (Nightingale 2000), and improved inter-organizational coordination and knowledge acquisition (Von Hippel and Von Krogh 2006).

While prior research has largely documented ICT-enabled evolutionary changes to innovation processes, this study views the innovation impacts of ICT in the context of observations about General Purpose Technologies (GPTs). GPTs are characterized by high fixed costs, low marginal costs of reproduction, and may be viewed as enabling technologies rather than directly shifting the production function of firms that use them; they also shift the invention possibility frontier of the economy, setting off waves of valuable co-invention (Bresnahan and Trajtenberg 1995, Jovanovic and Rousseau 2005). In brief, as GPTs evolve and advance, their use spreads throughout the economy across a wide range of

sectors to yield generalized productivity gains and pervasive innovation complementarities. The GPT nature of ICT has been widely acknowledged in the literature (Bresnahan and Greenstein 2001, Forman et al. 2005, Venturini et al. 2013). However, the empirical identification of innovation complementarities remains under-investigated. This is reflected in the call for more microstudies (Bresnahan and Trajtenberg 1995, Koutroumpis et al. 2017) that estimate the magnitude and nature of interactions between ICT and using sectors, as well as those that estimate how much of the gains from innovation registered in the using sectors are due to technological advances in the sectors themselves rather than from innovations in ICT incorporated in the sectors.

Prior empirical studies have analyzed patent citations to identify specific classes of ICT technologies that may be characterized as GPTs and shifts in modal invention trajectories facilitated by these technologies. For example, Hall and Trajtenberg (2004) identify individual patents as general-purpose based on their generality and association with rapidly evolving technology classes. Similarly, Koutroumpis et al. (2017) identify technology classes or fields that were adopted and cumulatively invented upon by other technology fields. The assumption underlying these approaches is that GPTs are best characterized as generative clusters of inventions with spillovers generated by the technological subfields indicating their broad applicability and pervasive impacts on invention. Both studies emphasize “*the rise of ICTs as invention machines*” (Koutroumpis et al. 2017) that play a critical role in the process of invention in diverse technology fields.

Our study advances this literature and is aligned with the notion that GPTs have pervasive impact. However, while we too use patent data, we focus on innovation complementarities across industries rather than across technology fields. Specifically, we estimate the magnitude and nature of complementarities between the ICT industry (that creates GPT technologies) and other industries (that use the technologies). Our analyses, therefore, respond to the call for more research on the nature of innovation complementarities between industries, including shifts in products, processes and competitive structures of the citing industries. The estimation of complementarities leads to two testable predictions in our study, a) that technological innovations of the ICT industry are driving shifts in technological innovations across diverse industries, (b) that ICT-intensive innovations impact the nature

of R&D and returns thereof in the citing industries. We also provide empirical insights into these innovation shifts and their implications for competitive dynamics in citing industries.

2.2 An ICT-Biased Shift in R&D

Our first testable prediction is borne out by anecdotal evidence and industry reports, which suggest that ICTs are becoming integral to the content of a range of industrial-age products from cars to books to automobiles (Yoo et al. 2012, Porter and Heppelmann 2015, Branstetter et al. 2019). This “digitization” of products has evolved over time with the exogenously evolving innovation trajectory of the ICT industry that counts its components as miniaturization of hardware, increased computing power, ubiquitous wireless communication, and low-cost storage media, amongst others (Porter and Heppelmann 2014). For example, consider the evolution of sensor technology and its transformational impact on the products and business models of automobile manufacturers over time. Traditional electromechanical sensors were standalone nodes that were specific to a context, measured a limited number of parameters, were costly to develop and implement, and as a result, were limited in their capability and scope. Manufacturers made little use of sensors or electronics. In general, products were stand-alone and mechanically controlled, and product development was largely limited to mechanical engineering. Not surprisingly, through the early nineties, product differentiation in the automobile sector centered on physical attributes (e.g. size, horsepower and aesthetics of automobiles). Manufacturers had a little to no access to data from their products and relied on downstream players in the value chain for feedback on product performance. Data collected during maintenance checks and repairs were used to develop insights on various components and discern performance.

In the late nineties, the ICT industry began high-volume deployment of wireless sensors while reducing the cost and energy per sensor as well as development and maintenance of such sensors. Examples of enabling initiatives include NASA Sensor Webs (2001), ZigBee Alliance (2002), Center for Embedded Network Sensing (2002), and Smart Dust, amongst others. The subsequent accelerated development of sensor technologies, including microelectromechanical systems, CMOS-based sensors and LED sensors along with advances in complementary technology areas such as CMOS-based semiconductor devices, networking protocols and energy storage/ generation technologies, helped

achieve the ICT industry's objective of widespread deployment of wireless sensors. The emergent "Internet-of-Things" is but a culmination of this decades-long effort of technological innovation.

The above-mentioned evolution of sensor technologies in the ICT industry has profoundly impacted technology development in the automobile industry. Modern sensors power automated tire pressure monitoring, accelerometers and high-speed actuators in airbags, on-board diagnostics modules, toll passes, theft recovery devices, and drive-by-wire systems that have replaced traditional mechanical control systems, amongst other functionalities in the modern automobile. Light detection and ranging (LIDAR) sensors that offer detection and control even in the complete absence of light yield even better and more consistent and reliable features such as blind spot monitoring, adaptive cruise control and braking, collision avoidance and pedestrian protection.

The impact of sensor technologies is but an illustration of how technological innovation in the ICT industry has significantly influenced the product design and innovation principles of the automotive industry. "Mechatronic features", which refer to the integration of mechanical systems with electronic and computational systems for provision of product functionality, are increasingly replacing mechanical features as the basis of competitive differentiation in modern automobiles. Indeed, industry observers contend that more than 70% of all innovations in the contemporary automotive sector are driven by ICT (Grimm 2003, Shorey 2011). To increase the pace of innovation, automakers are going beyond proprietary technologies to open up their operating systems (e.g. Ford OpenXC) to third-party developers to create varied apps for navigation, communication and entertainment as well as connect automobiles with mobile devices. In turn, they have several conduits to collect data, including "in use", for feedback and product modifications akin to software updates (Iyer and Venkatraman 2015). Not surprisingly, value creation and capture in the automobile industry today is increasingly realized through ICT capabilities enabled by inventions of the ICT industry (Grimm 2003, Frischkorn 2004).

2.3 The Performance Impacts of ICT-Enabled Innovations

Our second testable prediction is that ICT-enabled technological innovations positively impact returns to R&D investments in the citing industries and competitive dynamics thereof. Our prediction regarding impacts on R&D investments is borne out of prior research, which emphasizes three defining characteristics of ICT-enabled technological innovations. First, ICT enables separation of information

and logic from a physical artifact that embodies the information and executes the logic, thereby, significantly reducing the marginal costs of modifying or reprogramming products. For example, testing different levels of horsepower in traditional automotive engines required manufacturing multiple engines, each providing a different level of horsepower. However, today, engine manufacturers can easily alter the horsepower of a standard physical engine using software alone. The informational component of the product allows for real-time and emergent needs to be addressed in a predictive manner as well as for ongoing quality management that extends the useful life of the product. Consider that Tesla recently performed an over-the-air software update to improve the safety performance of its product. In brief, a modal shift to ICT-enabled innovation reduces the costs of prototyping, designing variability, modifying product designs and product updates and support.

Second, in addition to reducing the marginal costs of product design and reconfiguration, ICTs improve innovative efficiency by enabling cataloging, sharing and reuse of product components and functions and in turn, interoperability across products that are traditionally considered disparate. For example, the industrial applications and services on GE's Industrial Internet platform, Predix, which facilitates data management and analytics, visualization, asset optimization, and geospatial intelligence, amongst others, is used by a variety of industrial equipment manufacturers across diverse sectors ranging from aviation to energy. This modular product architecture, engendered by digitization, generates scale economies for innovation and significantly improves innovative efficiency of the firm.

Finally, ICTs are self-referential in that they leverage and requires an ecosystem of other ICTs, including digital devices, networks, and services to create value (Benkler 2006, Hanseth and Lyytinen 2010). For example, consider Nest Labs, which introduced a learning thermostat in 2011 that programmed itself for energy optimization. In contrast to an analog counterpart that is a standalone product, the digital thermostat is a networked product in that its value increases with the number of other products that it can connect to, personalization, and contextual awareness. Not surprisingly, in subsequent product versions, Nest created a platform, which linked a set of its own products in the home via a smartphone app and established an ecosystem involving major players such as Whirlpool, Mercedes-Benz, Fitbit, and others. The products of the latter set of companies had been digitized and connected with the Nest platform to create a broader digital ecosystem of hardware, software,

connectivity and services with many different business models. The example emphasizes that ICT-enabled innovations, in engendering self-referential and positive network externalities, accelerate innovative activity through “*a virtuous cycle of lowered entry barriers, decreased learning costs, and accelerated diffusion rates*” (Yoo et al. 2010).

The above defining characteristics of ICTs should improve, not only innovative efficiency, but also another key dimension of innovative performance, recombinant capabilities (Henderson and Clark 1990, Yayavaram and Ahuja 2008, Carnabuci and Operti 2013)². These capabilities include the ability to envision and create new technological combinations using technologies that have never been combined before - *recombinant creation* – and the ability to refine and reuse systematically known technological combinations to solve new problems and develop new applications - *recombinant reuse*. The separation of information from a physical artifact and allied sharing and reuse of product functions as well as interoperability across products is especially salient to the development of recombinant capabilities. As an illustration of how technologies of the ICT industry facilitate recombinant creation, consider again the example of GE. GE manufactures industrial machines such as power generating turbines, medical devices, diagnostic products, and commercial engines, amongst others. “Break-fix” service agreements associated with these industrial assets contributed to a bulk of the company’s revenues until 2011. In turn, the company owned several discrete technologies such as machine status notification systems, comparators and self-learning evidential reasoning systems that minimized discrepancy between actual output and desired output, and methods for analyzing error logs and other big data, all of which sought to detect machine failure and provide a quick and timely maintenance support in response. For example, patent number US 6591296 B1 automatically generated and sent an email message to a machine diagnostic center when a sensor detected a machine failure or predefined abnormality. Beginning 2011, the company began to leverage the exogenously evolving ICT trajectory

² Henderson and Clark (1990)), in their study of the photolithographic equipment industry, argue that incumbent industry leaders often lose their technological leadership due to their inability to link well-understood and established technologies in new ways. Yayavaram and Ahuja (2008)) show that semiconductor firms generate more impactful ideas when they recombine technological components from otherwise decoupled knowledge domains. Carnabuci and Operti (2013)) disentangle the conceptualization and operationalization of recombinant creation from recombinant reuse.

to accord its industrial machines an informational or “smart” component and connectivity. As a result, GE began to combine hitherto discrete technologies to create systems and methods for prognostics and increasing the efficiency of an entire ecosystem of machines. For instance, a set of filters, receptive of information on operational parameters of a machine, were combined with connectors to other machines and operating environments. These were further integrated with comparators and methods for analyses of big data to create a degradation monitoring system (patent number US 7571057 B2) that indicates degradation of the machine and an underlying cause.

The combination of these technologies also yielded a predictive condition monitoring system (US 9256224 B2) that forecasts the future operational condition of the machine. ICT technologies such as sensors, communication systems and protocols, and computing models and algorithms played a critical role in both these instances of recombinant creation. This shift in GE’s portfolio of technologies also facilitated a shift in the nature of its service contracts from “break-fix” to “real-time maintenance,” “preventive maintenance” and “outcome-based”.

ICT technologies are also critical to recombinant reuse. For example, the degradation monitoring system articulated above, although developed in the context of wind turbines, is applicable to a range of other industrial equipment of the firm. Such reuse of ICT technologies is an outcome of their unique properties discussed earlier, notably, data homogenization and reprogrammable functionality. These properties render disparate ICT technologies interoperable and scalable across business contexts.

These arguments emphasize that knowledge transfers from the ICT industry may be economically significant. That is, industries that incorporate technologies of the ICT industry to drive innovation will exhibit systematically better R&D outcomes than industries who do not.

McAfee and Brynjolfsson (2008) document accelerated competition since the mid-1990s that is coincident with a sharp increase in ICT investments in business processes. Indeed, the shift in competitive dynamics is most pronounced in sectors that have spent the most on ICT. We suggest that the competitive impacts of digitization of processes must extend to digitization of products too. A firm’s innovative capabilities are VRIN and in turn, contribute to its competitive (e.g. Henderson and Cockburn (1996) and Arora et al. (2009)). Therefore, we expect that industries that incorporate technologies of the ICT industry will exhibit systematically different competitive dynamics.

3. Data

Data Sources: Our sample, constructed at the industry-year level, is at the intersection of Compustat, Center for Research in Security Prices (CRSP), and patents data provided by the United States Patents and Trademark Office (USPTO). We retrieve accounting data for constituent firms in the industry from Compustat and market returns data from CRSP. We obtain patent application data from Patent Examination Research Dataset (PatEx)³. From this dataset, we extract patent bibliographic information such as application number, filing date, classification of the invention, examiner id, status, patent number, and issue date on all the patent applications. We also obtain information on average number of claims per application of a patent from the Patent Claims database, and patent family size – set of patent applications filed with different patenting authorities that refer to the same invention – from Foreign Priority data. These data have been used in prior research (e.g. Lemley and Sampat (2012)), and are available for the time period 1931-2017. We obtain data on patent citations from Google’s patent data to construct the citation network. To estimate the innovativeness of industries, it is imperative that we accurately match patent data from various sources of USPTO to firm level data obtained from CRSP or Compustat. We obtain these matched data from (Kogan et al. 2017), who identify CRSP firm identifiers for assignees of all utility patents granted by the USPTO between January 1926 and November 2010.

Since the pioneering work of Jaffe et al. (1993)), patent citations have been extensively used to study knowledge flows across time, technological fields, firm boundaries, strategic alliances, and social networks (for example, see Jaffe and Trajtenberg (2002), Peri (2005), Alcacer and Gittelman (2006), Gomes-Casseres et al. (2006)). Prior research, including seminal work by Griliches (1979) and Jaffe (1986), assume discrete relationships between industries in estimation of these knowledge flows. In turn, these studies conceptualize knowledge flows between a pair of industries as the technological distance between them that is estimated as one of: (a) extent of overlap in product fields between the industries (Verspagen 1997), (b) rates of cross-citations of patents between the industries (Jaffe et al.

³<https://www.uspto.gov/learning-and-resources/electronic-data-products/patent-examination-research-dataset-public-pair>

1993, Verspagen 1997), or (c) cross-classification of patents by industry of production and industry of use (Scherer 1982, Kortum and Putnam 1997)⁴.

In contrast to this view of prior research, we adopt a network approach to estimating knowledge flows between industries. Such an approach to estimating knowledge flows assumes that in addition to originating and receiving knowledge, an industry may also act as a conduit for knowledge transfer between industries (Ahuja 2000). Specifically, an industry's knowledge linkages with another industry (direct ties) provide it with access not just to the knowledge held by the latter but also to the knowledge held by the latter's knowledge partners (indirect ties). These indirect ties increase the quantum and diversity of knowledge flows to the industry and are especially salient to estimating knowledge transfer from networked and self-referential digital innovations.

The first step in constructing such a network of inter-sectoral knowledge flows involves the assignment of patents to an industry. Hence, we assign each patent in our sample to the primary 4-digit Standard Industrial Classification (SIC) code of the assignee and construct inter-sectoral citation networks for each year in our sample period. Specifically, for each year, we construct a directed and weighted network where the nodes or vertices, are identified by their 4-digit SICs. The edges that connect these industries are directed and weighted – the direction of the edge is from the cited to citing industry, representing the direction of technology impact or knowledge (innovation) flow; the weight of the edge is the 'citation intensity'. We define citation intensity for pair of industries A and B as the number of citations made by the patents of industry A to the patents granted over the past five years in industry B . Specifically, say, in Year y , the number of citations made by Industry A to patents granted to Industry B during the Years $(y - 5)$ to $(y - 1)$ is p . If $p > 0$, we create an edge from Industry B to Industry A with a weight of p for measurement Year y ; $p = 0$ implies no edge between industries A and B . The five-year period used in computing citation intensity is broadly consistent with prior research, which suggests that technology cycles measured by the duration of rents extracted from R&D

⁴ Industries of use (including final consumption) are classified as one of two "general use" categories - (1) use proportional to the origin industry's normal customer sales distribution; or (2) ubiquitous use throughout the industrial economy. This 'technology flows' table is then used to reweight R&D data by line of business into measures of "original" and "imported" R&D, assuming that the knowledge flows to industry A from industry B is reflected in the proportion of B 's patents deemed to be used for industry A .

are approximately five years in most industries (Lev and Sougiannis 1996, Gu 2005). Going forward, we will use the words *industries*, *sectors*, *nodes*, and *vertices* synonymously to refer to industries; we will use the words *ties*, *links* or *edges* to refer to connections between industries in our citation network.

Although our patent data begins in 1926, we construct the inter-sectoral citation network beginning 1981 to ensure the quality of R&D expenditure data. The accounting treatment of reporting R&D expenses was standardized in 1975 (Financial Accounting Standards Board Statement no. 2). Consistent with prior research (Cohen et al. 2013, Hirshleifer et al. 2013), we estimate R&D capital (that is the denominator for our dependent measure of innovative efficiency) beginning 1979 to allow for a full five-year period with reliable R&D expenditure data. In turn, the two-year patent application-grant lag renders 1981 our first year of estimation of innovative efficiency that is normalized by five-year R&D capital. We could obtain Compustat or CRSP firm identifiers for the patent assignees until 2010. Thus, our final sample of patent data includes 1,314,962 patents granted by the USPTO from 1981 to 2010, yielding 30 inter-sectoral citation networks across 306 industries (identified by their 4-digit SIC codes).

We describe innovation outcomes and key network measures used in our analyses below.

4. Variable Measures

4.1 Dependent Variables – Innovation Outcomes and Industry Fundamentals

We aim to assess the economic significance of knowledge transfers from central industries to other industries in the inter-sectoral citation network. To this end, we analyze the impact of the closeness of an industry to the central industries on its innovation outcomes and competitive dynamics. Below, we describe the measures of innovation outcomes and industry competitiveness.

4.1.1 Innovative Efficiency

The first innovation outcome that we estimate is innovative efficiency or patents granted per unit of R&D investment. Innovative efficiency reflects the efficiency of R&D investments and is different from the patenting intensity that adjusts patenting output for the size or scale of operations of a firm. We estimate the innovative efficiency of an industry as the mean innovative efficiency across its constituent firms. Following prior research (Cohen et al. 2013, Hirshleifer et al. 2013), we define the innovative efficiency of a firm in measurement year y as the ratio of the number of patents granted to the firm in year y to the R&D capital — five-year cumulative R&D expenditures assuming an annual depreciation

rate of 20% — ending in year ($y - 2$). We allow a two-year gap between R&D capital and patents granted as it takes, on average, two years for the USPTO to grant a patent (Hall et al. 2001). In unreported results, we find that the use of two-year lagged R&D expenses instead of R&D capital as the denominator in our measure of innovative efficiency generates similar results.

4.1.2 Recombinant Innovation Indices

As noted earlier, an important element of innovativeness of firms is their ability to recombine existing technologies (Henderson and Clark 1990, Yayavaram and Ahuja 2008). Following prior research (Galunic and Rodan 1998, Yayavaram and Ahuja 2008, Carnabuci and Operti 2013), we posit two measures of recombinant capabilities, notably, recombinant creation and reuse — the former refers to combining technologies that have never been combined before while the latter refers to improving existing combinations of technologies to solve new problems, find new applications, and identify new contexts where previous knowledge can be used. We first estimate these recombinant capabilities at the firm-level (as articulated in Table S1 in Online Appendix), and then compute recombinant indices of an industry by taking the average of the corresponding indices of all firms in that industry.

4.1.3 Industry Fundamentals

Following McAfee and Brynjolfsson (2008)), we measure the competitiveness of an industry using not just the average industry performance but also the volatility and spread in such performance. Specifically, we estimate mean cash flow, cash flow volatility and cash flow spread for the sample industries. Consistent with prior research (e.g., Hirshleifer et al. (2013))), we estimate cash flow as income before extraordinary items less total accruals (changes in current assets plus changes in short-term debt and minus changes in cash, changes in current liabilities, and depreciation expenses) divided by average total assets. Cash flow volatility is estimated as the three-year standard deviation in cash flow. Finally, we estimate the spread in cash flow for a particular industry-year as the difference in cash flow between the firms performing at the 25th percentile and the firms performing at the 75th percentile for the focal year in that industry.

4.2 Key Network Measures

4.2.1 Innovation Centrality of an Industry

The accurate assessment of an industry's position as a knowledge contributor in the citation network requires use of a network centrality measure that captures its overall contribution with respect to knowledge flows. There exist several centrality measures like degree, closeness, betweenness, and eigenvector centrality. The closeness centrality of an industry measures the industry's access to the technologies of other industries but does not capture the extent of contribution of the industry in terms of innovation. The betweenness centrality measures the strength of an industry to act as a conduit of knowledge transfer between two distinct groups of industries but just like closeness it does not measure the extent of contribution either. Out-degree on the other hand does capture the extent of knowledge transfer from the focal industry through total number of direct citations, however it does not capture the innovation diffusion beyond the immediate industries. We note that knowledge flows in the network may be characterized as a walk (an unrestricted sequence of nodes and links), a trail (a sequence in which no link is repeated), a path (a sequence in which no link or node is repeated), or a geodesic path (the shortest path between two nodes). Further, knowledge along the network may spread by transference or replication (Borgatti 2005). Eigenvector centrality captures the innovation diffusion beyond the immediate industries, and also accounts for the nature of knowledge flows discussed above.

More precisely, *Eigenvector centrality*, uses count of direct citations as well as the innovation impact of the citing industries to measure how important a node is in the citation network. That is, industries have high eigenvector centrality if they are connected to other industries that are themselves more cited and hence, more central. Therefore, eigenvector centrality is the most appropriate measure of innovation centrality. We compute the eigenvector centrality for each industry using the citation network constructed for every year in our sample period. The finding of high centrality for the ICT industries as well as growth in such centrality over time would validate our prediction that technological innovations of the ICT industry are increasingly driving innovations across other industries.

4.2.2 Closeness to Central Industries

We estimate the extent of knowledge flow between any two industries (or nodes in the network) using a network statistic known as closeness. Consider any two industries, A and B, in the citation network.

If there is a direct edge from A to B, we define the closeness of B to A as the weight of that edge. Since the edge weight is the same as citation intensity, the closeness of B to A, in this case, proxies for the extent of knowledge transfer from A to B. However, if there is no direct edge from A to B, the computation of closeness is not straightforward. In such case, computing the longest path from A to B may seem to serve the purpose of estimating knowledge transfer. However, that will lead to unbounded closeness when the network has cycles. Therefore, we take the following approach to estimate the closeness of B to A. First, we take the inverse of the weight of the edge between a pair of nodes in the network as the distance between the pair. We then compute the distance of each path from A to B by summing the distances of each edge that falls along the path. Of all the paths, we identify the one with the shortest distance. The inverse of the path length of this shortest path estimates the closeness of B to A. Furthermore, when there exists no path from A to B, we take the closeness of B to A as zero. The closeness of B to A, therefore, captures the extent of knowledge consumption by B from A. Further, we define the closeness of a non-ICT industry from ICT industries as the average closeness of that non-ICT industry from all the constituent ICT industries, and we denote this measure as ICT-Closeness.

The significance of knowledge transfer involves an assessment of the effect of research performed in one economic unit in *improving* technology in a second economic unit. Therefore, if indeed there are significant knowledge flows from the ICT industry across diverse industries, merely demonstrating the closeness of these industries to the ICT industry would not suffice. We must demonstrate that such closeness is economically significant as manifest in the positive impact of closeness to the ICT industry on innovation outcomes of the citing industries.

5. Model & Estimation

We assess the impact of the closeness of non-ICT industries to the ICT industries in an inter-sectoral citation network on the former's innovation outcomes (innovation efficiency and recombinant capabilities) using the following model:

$$Inn_Outcome_{k,i,y} = \beta_0 + \beta_1 ICT_Closeness_{i,y-1} + \sum_m \beta_m X_{m,i,y-1} + \delta_i + \delta_y + \varepsilon_{i,y},$$

where the explanatory variable $Inn_Outcome_{k,i,y}$ represents one of the three dimensions of innovation outcome of the non-ICT industry i in year y , where k identifies innovative efficiency, recombinant reuse, or recombinant creation; $ICT_Closeness_{i,y-1}$, the main variable of interest, represents ICT-Closeness

of non-ICT industry i to the ICT industry in year $y - 1$; $X_{m,i,y-1}$ represents various measures of non-ICT industry i in year $y - 1$ that are used as controls. Our controls include measures like IT Capital, R&D intensity, and marketing intensity of the average firm in the industry, as well as other time-varying, industry-specific factors that may influence innovation outcomes. These control measures along with the key independent and dependent variables are detailed in Table S1 in Online Appendix. Finally, δ_i and δ_y capture the industry and year fixed effects. We use seemingly unrelated regression (SUR) to estimate these three measures of innovation outcomes since a separate estimation would ignore potential contemporaneous correlation amongst the errors. All models control for time-invariant industry and year fixed effects with robust standard errors.

Insert Table S1 here

5.1 Identification

A potential concern with above specification is that estimates may be subject to endogeneity biases. For instance, non-ICT industries, in expectation of superior innovation outcomes, may systematically create innovations that bring them closer to ICT industries. Therefore, we need to control for unobserved heterogeneity that impacts the self-selected choice of firms in an industry to cite patents of another industry, that also impact innovation outcomes. These unobservable factors include network processes amongst other industries that impact the ICT-Closeness of the focal non-ICT industry. We adopt the Instrumented Network approach (Boehmke et al. 2016) to address these endogeneity concerns in our estimation. This approach broadly involves the following steps and we elaborate on each of these steps subsequently. First, we estimate the network formation using an instrument variable that can induce exogenous variation in the dyadic relationship (or tie formation between any two nodes). Second, we simulate a network using the estimated model and calculate the network measure of interest from the simulated network. These two steps are similar to the first stage estimation of the traditional 2SLS model, except that these steps also account for the network formation. In the final step, we use this network measure to estimate its impact on the dependent variable.

Two important challenges need to be addressed in adopting the Instrumented Network approach. First, the network simulation model should have inferential characteristics to allow for randomization-

based hypothesis testing. This property is essential to reject the null hypothesis of the weak instrument or significance bias of the instrument variable estimator relative to the naïve estimator. Second, the instrument variable must satisfy the exclusion restriction, that is, it must affect the dependent variable (innovation outcomes) only through the network measure (ICT-Closeness). In the subsequent sections, we describe our network simulation model, and provide details on our choice of the instrument variable.

Network Simulation:

In the instrumented network approach introduced in Boehmke et al. (2016)) a linear regression model is used to generate dyadic relationships in the network. However, a linear model does not entirely capture the non-linear interconnectedness of nodes in a given network. Further, Smith et al. (2016)) use temporal network autocorrelation method to model dyadic relationships in the instrumented network approach. However, this method requires one to manually define all network dependencies that are essential for network formation, which are complex to include. Further, robustness of this method for randomization-based hypothesis testing is unknown (Silk et al. 2017). Hence, we propose an instrumented network approach that uses Exponential-family Random Graph Models (ERGMs) to model dyadic relationships and simulate the citation network. However, as our network is weighted, we use valued-ERGM as opposed to the commonly used binary ERGM model.

Exponential-family Random Graph Models are generative models for networks, which assume that the network space of interest can be represented by an exponential family specified by sufficient statistics (Morris et al. 2008). These sufficient statistics are considered to be mutually exclusive and completely exhaustive in their representation of network characteristics that underlie network formation. This modeling approach allows us to include structural effects like distance and group membership, amongst others that are suited to the study and design of social systems (Silk et al. 2017). Further, Cranmer and Desmarais (2011)) note that ERGM models can be used for inference, which is essential to rule out endogeneity issues. Prior research (Snijders et al. 2006, Robins et al. 2009, Lusher et al. 2013, Wang et al. 2013) in management evaluates the appropriateness of the ERGM approach in understanding network formation. In our study, we are interested in simulating a weighted and directed network, hence we use valued ERGM.

We draw on prior research (Lemley and Sampat 2012, Sampat and Williams 2019) to use

examiner experience as a source of variation in citations across patents. Several studies (Merges 1999, Lemley and Sampat 2008) assert that the assignment of examiners to patent applications is plausibly random given the application year and technology type. Further, while examiners are charged with a uniform mandate – grant patents to eligible, novel, non-obvious and useful inventions – prior research (Sampat and Lemley 2010, Lemley and Sampat 2012) also documents the significant discretion available to patent examiners in granting patents and reducing claims thereof. Thus, examiners play a potentially critical role in directing knowledge flows across inventions. We leverage the specificity of an examiner to a patent, heterogeneity across examiners, and the quasi-random assignment of patent applications to examiners as a source of variation in the patent citations.

While the institutional details of the patent review process and prior research (Sampat and Williams 2019, Melero et al. 2020) suggest that assignment of applications to examiner types is plausibly orthogonal to drivers of innovation outcomes at the patent level, our analysis is conducted at the industry level. Hence, we provide empirical support for the validity of examiner experience as an instrument variable at the industry level in two steps. First, we use patent application characteristics that are fixed at the time of examiner assignment to predict citation probability among industries. We subsequently test whether our main variable of interest, ICT-Closeness, estimated for the citation network predicted in the previous step, relates to the experience of assigned examiners. The absence of a relation between predicted ICT-Closeness and examiner experience is suggestive of random assignment of examiner at the industry level.

Additionally, we examine the relationship between examiner experience and predicted innovation outcomes (based on observable industry attributes). In particular, we regress innovation outcomes on determinants of innovation performance detailed in prior research. We then regress this predicted innovation outcome on examiner experience. The absence of a significant relation between these variables suggests that industries more likely to have superior innovation outcomes are not systematically assigned to examiners with less or more experience. We elaborate on both these exogeneity checks in Section 6.2.

We use examiner experience as a node attribute in simulating the network through ERGM. ERGM estimates the effect of different attributes in network formation through a network statistic

computed as $\sum_{i \in N} \sum_{j \in N} x_i y_{i,j}$, where N is the set of all nodes in the network, i and j are the indices of the nodes, x_i is the value of one such attribute of node i , and $y_{i,j}$ is the weight of the directed edge from node i to node j , which in our context, is the count of citations from industry j to industry i . Given that examiner characteristics influence citations to other patents from the focal patents, we use the network statistic $\sum_{i \in N} \sum_{j \in N} Experience_i y_{j,i}$ to capture the impact of examiner experience in tie formation. Therefore, the ERGM specification of the citations network is:

$$\theta_1 \sum_{i \in N} y_{i,j} + \theta_2 \sum_{i \in N} \sum_{j \in N} Experience_i y_{j,i}.$$

Using this model specification, we estimate the citation network for each year between 1981 and 2010. The distribution of the edge weights is specified to be binomial.

As a next step in our analyses, we use the model estimated above to simulate the citation network for each year between 1981 and 2010, separately, using Markov Chain Monte Carlo (MCMC) method. While the ERGM approach uses the coefficients in the model specification to define the probability distribution of network configurations across all networks of the same size, the simulation using MCMC approach draws a network from this distribution that is more likely to “resemble” the observed network. To confirm that our simulated network is not degenerate (Krivitsky 2012), we compare various network statistics (beyond the ones that are used in modeling the network) between the actual and simulated networks to confirm that our simulated network is not degenerate. In particular, we compare both the degree distribution (a network statistic included in the model) and the R&D weighted centralities (a network statistic not included in the model) of the simulated networks with those of the actual networks. Comparative statics confirm that our networks are not degenerate but rather, are close to the real network. The *sufficient* statistics used in ERGM for simulating the network involve characteristics of all the industries in the economy. The ERGM results show that these characteristics are significant in simulating the network. We use the simulated networks for all years to estimate the closeness of each of the non-ICT industries to the ICT industries

Finally, we regress the pertinent innovation performance metrics of non-ICT industries in year y on the predicted ICT-Closeness of those industries in year $y - 1$ while controlling for a range of time-

variant industry-level controls with industry and year fixed effects and robust standard errors. This estimation of performance corresponds to the second stage in a traditional 2SLS model.

6. Results

6.1 Network Structure of Inter-Sectoral Citations

The structure of the citation network provides insights into the relative positions of constituent industries, including their centralities, and skewness of knowledge flows within the network. Figure 1 presents the visualization of the citation network for our sample of industries in 1997. For reasons of brevity, we present the network structure for the year 1997 alone. Although we do not present the network structure for all years in our sample period, we verify and note that the results that follow do not vary qualitatively across years.

In Figure 1, each node in the network represents an industry identified by its 2-digit SIC code⁵. The size of the node represents the out-degree of the node - the greater the number of citations that an industry receives, the greater is the size of the node. Figure 1 indicates the skewed out-degree of our sample industries in 1997, where a few industries receive a bulk of the citations.

Insert Figure 1 here

6.1.1 Centrality of the ICT Industries in the Citation Network

As the next step in our analyses, we identify the central industries in our network. Specifically, we identify the five most central industries each year from 1976 through 2010. Table S2 in Online Appendix lists these central industries by year.

Insert Table S2 here

We further plot the centralities of all our sample industries in Figure 2. Here, each line traces the eigenvector centrality of an industry in our sample period from 1976 through 2010. The centrality score for each industry is normalized to a 0-1 scale within a year. The lines that correspond to the five ICT industries, identified by industry codes 35, 36, 38, 48 and 73, are colored in orange.

⁵We use 2-digit SICs to obtain broader classification of industries and more meaningful visualization of the network properties in Figures 1 & 2. However, rest of the analyses are conducted at the level of 4-digit SICs.

Figure 2, in conjunction with Table S2, suggests that the innovation centrality of a majority of the ICT industries (SICs 35, 36, 73) has either increased significantly or remained high in the early nineties. The nineties marked the start of a period of significant innovation in the ICT industry, notably, the advent of the Internet and enterprise software applications that became fundamental to business. A rich body of prior research (Barua et al. 1995, Hitt and Brynjolfsson 1996, McAfee and Brynjolfsson 2008) has documented and demonstrated that these innovations in the ICT industry were accompanied by a surge in corporate investments in ICT and an allied increase in productivity and shift in competitive dynamics of ICT-centric firms and industries. Figure 2 suggests that in addition to these documented impacts, innovations in the ICT industry were accompanied by a significant discontinuity in the nature of R&D across industries. The biggest contributors to this discontinuity seem to be manufacturers of computer equipment (SIC 35), electrical and electronic equipment (SIC 36), and the software services sector (SIC 73). The centrality of the computer equipment manufacturing industry (SIC 35), while rapidly increasing through the nineties, began to decline slightly relative to SICs 36 and 73 in the late 2000, likely because of disruptive exogenous technological shocks such as the advent of wireless handheld devices and on-demand computing that reduced investment in costly IT assets. The centrality of the electrical and electronic equipment manufacturing industry (SIC 36), mainly, semiconductors, remained significantly high for most of the period. Interestingly, through the nineties, sales to the computer equipment manufacturing industry accounted for about 50 percent of sales of the electrical and electronic equipment manufacturing industry (Jorgenson and Stiroh 2000). Yet, the centrality of the latter did not follow the decline of that of the computer equipment industry, suggesting that technological innovations of the electrical and electronic equipment manufacturing industry were absorbed by other industries. At the end of our sample period in 2010, the five industries with the highest centrality were the five ICT industries.

The discontinuity in R&D in the late nineties is manifest, not only in the significant increase in the centrality of the ICT sector but also the contemporaneous decline in the centrality of certain other industries. For instance, the centralities of industries, identified by SIC codes 13, 26, 28, 29, 37, and 99, colored in green, were relatively high until the mid-nineties but began to decline thereafter. The decline in centrality in the mid-nineties is especially pronounced in the case of the Chemical and Allied Products

industry (SIC 28) and Transportation Equipment (SIC 37). The rest of the industries, colored in blue, have relatively low innovation centrality throughout the sample time period.

Insert Figures 2 & 3 here

Figure 3 compares the weighted average innovation centrality of the five ICT industries with that of the non-ICT industries over time. We use the total R&D capital for an industry in the measurement year as its weight while estimating the weighted average centrality. Consistent with Figure 2, we observe that the weighted average centrality of the ICT industries increases significantly in the early nineties and continues to remain higher than that of the non-ICT industries over the rest of the sample time period. In contrast, the weighted average centrality of the non-ICT industries that is comparable if not higher throughout eighties is significantly low compared to that of the ICT industries in the latter half of the sample time period. The higher centrality of ICT industries suggests that the number of industries citing patents created by the ICT industries witnessed a sharp increase in the early nineties that is sustained until the end of the sample period.

6.1.2 Closeness to the ICT Industries in the Citation Network

In the previous section, we identified the ICT industries as most central industry in the inter-sectoral citation network. In this section, we examine the extent of technology transfers from the ICT industries to non-ICT industries by estimating the closeness of the latter to the ICT industries in the network space. Specifically, we define the closeness of a non-ICT industry to the ICT industries as the average closeness of the focal industry to the different constituent ICT industries. The trendline in blue color in Figure 4 represent this average closeness of non-ICT industries over sample period. This confirms that the non-ICT industries are moving closer to ICT industries in the network space, suggesting that these industries are increasingly incorporating technologies created by the ICT industries in their innovations.

However, prior research (Graham and Mowery 2003, Hunt and Bessen 2004) points to regulatory changes in the patentability of technologies in the early nineties that resulted in a sharp increase in the stock of software patents of firms that are part of a larger class of ICT patents. It is possible that the observed closeness of diverse industries to the ICT industry is an outcome of this growing stock of ICT patents that are more likely to cite technologies of the ICT industry. Therefore, in order to verify that

the observed increase in closeness to the ICT industry is indeed a result of ICT-induced shifts in the nature of R&D rather than the above mentioned regulatory changes, we construct separate citation networks for ICT and non-ICT patents and differentially assess the closeness to the ICT industries over time in both these citation networks. If indeed there is a systematic shift in the nature of R&D in diverse sectors towards more ICT-centric technologies, we should observe increasing closeness to the ICT industry in both networks.

We first divide our sample of patents into two categories—ICT patents and non-ICT patents. We use the IPC classes identified in Corrocher et al. (2007)) to classify patents as ICT patents. Corrocher et al. (2007)) use patent abstracts to detect important applications in the ICT field by selecting the most frequent sequential triples of words that identify technological applications in the field. Their method yields a set of IPC classes related to these applications that are broader than those usually considered. We consider a patent as an ICT patent when, at least, one of the IPC classes assigned to the patent identifies ICT patents as defined in Corrocher et al. (2007)). Thereafter, we create separate networks using ICT and non-ICT patents, and estimate closeness to the ICT industry in these two networks over time. Further, we obtain the average proportion of ICT patents for non-ICT industries for the same time period.

Figure 4 affirms that the proportion of ICT patents of non-ICT industries is increasing steadily over time (trend line colored in black). Our result for the increase in the closeness of non-ICT industries to the ICT industries holds in both citation networks (trendlines in red and yellow). That is, the observed significant increase in the ICT-centricity of innovation in our sample industries over the sample time period is not entirely driven by the proliferation of software patents, documented in prior research but rather, reflects a fundamental shift in the nature of R&D towards more ICT-centric innovation. The results for the significant increase in the centrality of the ICT industry over the sample period and the concomitant closeness of the non-ICT industries to the ICT industry strongly support our view that technological innovations of the ICT industry are driving technological innovations across diverse industries.

Insert Figure 4 here

6.2 Impact of Technology Transfers from ICT Industries

In addition to demonstrating the closeness of non-ICT industries to the ICT industry, we must demonstrate that such closeness is economically significant as manifest in its positive impact on innovation outcomes of the citing industries. In doing so, we adopted the Instrumented Network approach discussed above to correct for endogeneity of network formation.

As the first step, we estimate the citation network using examiner experience as a covariate in the ERGM specification. We validate the exogeneity of examiner experience in two ways. Following prior research (Sampat and Williams 2019), we first assess whether there is evidence that patent applications that are close to the ICT industry are systematically assigned to experienced versus inexperienced examiners. Towards this end, we first predict the likelihood of the patent portfolio of industries being close to the ICT industry using two patent application characteristics – number of patent claims and patent family size that are fixed at the time of examiner assignment. Specifically, we average the values of these characteristics across patents issued in a particular industry-year and use it as a node attribute in the ERGM specification. An F-test of joint significance reveals that both characteristics have a significant effect in predicting the citation network. Next, we obtain the ICT-Closeness of non-ICT industries in the predicted citation network for each year and regress it on average examiner experience for the industry-year. We obtain a coefficient of 0.015 ($p = 0.874, SE = 0.094$), suggesting no evidence of a systematic relationship between the two variables. We conclude that there is no evidence of differential assignment of applications across examiner types at the industry level with respect to our main variable of interest (ICT-Closeness). This finding also corroborates with Figure S1 in the Online Appendix, which indicates no visual relationship between predicted ICT-closeness (predicted by the patent application characteristics) and the instrument, average examiner experience.

Next, we assess whether on average, industries with superior innovation outcomes are systematically assigned to experienced versus inexperienced examiners. We obtain a coefficient of 0.007 ($p = 0.965, SE = 0.164$) for the regression of predicted innovation performance on average examiner experience for the industry-year, indicating there is no systematic relationship between the two variables. Thus, there appears to be no evidence that industries with characteristics that render them

more likely to have superior innovation outcomes are systematically assigned to examiners with less or more experience. The above two analyses conclude that examiner experience satisfies the exclusion restriction conditions. Therefore, we use it as a node attribute in simulating the network through ERGM.

Insert Figure S1 here

For the sake of brevity, we present only the model estimates for the citation network corresponding to the year 1997 in Table 1. The network model estimates across other years in our sample time period are qualitatively similar. We find that the coefficient of examiners' experience is largely significant across years, and the AIC and BIC metrics, on average, have a lower value compared to a model specification when examiner experience is not included, further validating the appropriateness of including this variable in the ERGM specification.

Insert Table 1 here

As the next step, we simulate a network for each year using the model estimated above. Further, we ensure that the simulated network is not degenerate by comparing its network statistics with that of the actual network. Specifically, we compare the R&D capital-weighted eigenvector centralities as well as the degree distribution between the simulated and the actual networks. Figure 5 presents the average R&D capital-weighted eigenvector centralities of the non-ICT and the ICT industries using the simulated network over the sample period. We observe that the trend in R&D weighted centralities of the simulated and actual network (presented in Figure 3) are similar. Further, Figures 6a and 6b present the outdegree distribution of the actual and simulated network respectively for the year 1997. For reasons of brevity, we include outdegree distribution of the year 1997 only. As we observe in Figure 6, the degree distribution of the actual and the simulated networks are similar. These observations confirm that our simulated networks are not degenerate but are rather, close to the actual networks.

Insert Figures 5 and 6 here

Thereafter, we regress the performance metrics of non-ICT industries on the *ICT-Closeness* computed from the simulated network along with other control variables, and industry and year fixed effects; industry represented at the level of 4-digit SICs. The outcomes of these regressions are

presented in the following two subsections. Summary statistics of all the independent and dependent variables are included in Table 2. The correlation matrix of all independent variables is presented in Table S3 in Online Appendix.

Insert Tables 2 and S3 here

6.2.1 The Impact on Innovation Outcomes

For easier interpretation, we standardize all independent variables in our regressions. Furthermore, as we explained earlier, all our estimates are endogeneity corrected. The results of our analyses with regard to the impact of closeness from the ICT industry on innovation outcomes are presented in Table 3. Models I-III present the results for innovation outcomes - innovative efficiency, recombinant reuse, and recombinant creation, respectively. Our use of SUR is consistent with results of the Breusch-Pagan test of independence of equations, which finds that correlation among the residuals of the equations for innovative efficiency, recombinant creation, and recombinant reuse is significant ($p < 0.01$).

Insert Table 3 here

The coefficient for closeness to the ICT industry is significant for innovative efficiency and both recombinant creation and recombinant reuse. The results in Model I suggest that a unit standard deviation increase in the closeness of an industry to the ICT industry yields an increase of 10.2% increase in the innovative efficiency or number of patents per R&D dollar. While closeness is equal to the number of citations for industries that are connected to the ICT industry through direct citations, it is difficult to precisely translate closeness to the number of citations for industries that are connected to the ICT industry through indirect citations. Since the standard deviation of closeness for the industries in our sample is 1.26, a unit standard deviation increase in closeness is equivalent to having 1.26 more citations to the ICT industries for those industries that are already connected to the ICT industries directly; 48% of non-ICT industries are directly connected to the ICT industries in our sample period.

The results in Model II indicate that a unit standard deviation increase in the closeness of an industry to the ICT industry increases recombinant reuse by 0.15 units for that industry. Given that in our sample the mean industry's recombinant reuse is 0.31, this increase of 0.15 units represents a significant increase of over 48.39% in this capability. The results in Model III indicate that a unit

increase in the closeness of an industry to the ICT industry increases recombinant creation by 0.03 units for that industry. In our sample, the mean industry's recombinant creation is 0.33. Therefore, the increase of 0.03 units indicates a significant increase of over 9.09% in this capability.

Our results provide insights into the shifts in R&D outcomes that are triggered by technologies of the ICT industry. Prior research points to different pathways to achieving these outcomes. For example, Carnabuci and Operti (2013) posit that having an integrated intra-organizational network of inventors enhances a firm's ability to innovate through recombinant reuse by providing the inventors a stronger sense of identification with the firm and creating more stringent cognitive and normative schemas for problem definition and solution. However, such integration reduces the ability of the firm to innovate through recombinant creation. Similarly, a more diverse knowledge base hinders recombinant reuse but improves its ability to innovate through recombinant creation.

We find that ICT-enabled innovations enhance both recombinant creation and recombinant reuse. However, its impact on recombinant reuse (48.39% increase in capabilities) is greater than that on recombinant creation (9.09% increase in capabilities). Therefore, the characteristics of ICTs, notably, separation of information from the physical artifact in which the latter is embedded, interoperability of components and functions across traditionally disparate products, and self-referential nature, seem to be very valuable in applying existing technologies to new business contexts and application domains. Creation of new technologies, while impacted by ICTs, are likely still driven by other domain-specific factors such as quantum and diversity of domain knowledge.

6.2.2 The Impact on Competitive Dynamics

Models I-III in Table 4 present the results for the impact of *ICT-Closeness* on measures of industry competitive dynamics - cash flow, cash flow spread, and cash flow volatility respectively, for non-ICT industries. We do not find any significant impact of the *ICT-Closeness* on cash flow. McAfee and Brynjolfsson (2008)) suggest similar competition dynamics in IT-intensive industries. However, the impact of *ICT-Closeness* on both cash flow spread and cash flow volatility is significant. The results in Model II indicate that a unit increase in the closeness of an industry to the ICT industry increases its cash flow spread by 0.01 units. The mean industry cash flow spread is 0.18. This increase is nearly 5.56% for the mean industry. The results in Model III indicate that a unit increase in the closeness of

an industry to the ICT industry increases cash flow volatility by 0.001 units for that industry. The mean industry cash flow volatility is 0.02. This increase is nearly 5% for the mean industry. The results suggest that an industry's closeness to the ICT industry, while not altering the mean performance of the industry, significantly alters variance around the mean. Stated otherwise, digital innovations significantly alter competitive dynamics in an industry, including performance differences between leaders and laggards and volatility.

Insert Table 4 here

6.3 Mechanisms Underlying Impact

The results in the previous section suggest that the effects of the ICT industry on innovation are pervasive, and closeness of a non-ICT industry to the ICT industry in the innovation network space positively impacts innovation outcomes. However, thus far, we have not made progress on the more fundamental question of *how* closeness to the ICT industry shifts innovation trajectories of the focal industry to impact its innovation outcomes. In this section, we address this question.

Closeness to the ICT industry may matter because it results in greater integration of digital technologies, including *smart* and *connectivity* components⁶, with *physical* components (mechanical and electrical parts) that characterize traditional products (Porter and Heppelmann 2015). In turn, we should observe that the *ICT-Closeness* has a positive impact on the stock of digital technologies. We test this prediction by regressing the proportion of ICT patents of a non-ICT industry on its closeness to the ICT industry. We find that the coefficient of closeness is positive and significant ($p < 0.01$), implying that the closeness of a non-ICT industry to the ICT industry indeed results in an increase in stock of digital technologies in the focal industry. The results of this regression are presented in Model I of Table 5.

The positive impact of ICT-Closeness of an industry on its innovation outcomes may be explained, not just in terms of quantum of digital technologies, but greater complementarities among all

⁶ Examples of smart components include sensors, microprocessors, data storage, controls, software, embedded operating systems, and digital user interfaces. Examples of connectivity components include ports, antennae, protocols, and networks that enable communication between the product and the product cloud, which runs on remote servers and contains the product's external operating system.

the technologies of the focal industry. Prior research finds that digitization enables cataloging, sharing and reuse of product components and functions, and in turn, interoperability across products that are traditionally considered disparate. Digital technologies are also self-referential in that they leverage and require an ecosystem of digital technologies, including digital devices, networks, and services to create value (Benkler 2006, Hanseth and Lyytinen 2010). We should, therefore, observe that the greater the closeness of a non-ICT industry to the ICT industry, the greater is the complementarity of its patents. To test this hypothesis, we first obtain a semantic representation of each patent in our sample using its title and abstract based on *Doc2Vec* algorithm (Le and Mikolov 2014). As document vectors represent the semantic meaning, a higher similarity score between patents corresponds to a larger complementarity among their innovations. Hence, we compute the pair-wise similarities of patents granted in the focal year with those granted in the previous year, and use their average to represent the industry's 1-year complementarity in innovations for that year. Similarly, we compute an industry's 5-year complementarity in innovations by looking at the pair-wise similarities of patents granted in the focal year with those granted in the previous 5 years. We then regress these similarity measures of a non-ICT industry with its closeness to the ICT industry. Models II & III in Table 5 represent the results corresponding to the 1-year and 5-year similarity measures respectively. We find that the coefficient of ICT-Closeness is positive and significant ($p < 0.01$) in both cases, confirming that the closer a non-ICT industry is to the ICT industry, the greater is the complementarity. These results affirm that using sectors leverage technologies of the ICT industry to create complementary technologies in the industry, potentially, improving innovation outcomes.

Prior research (McAfee and Brynjolfsson 2008) finds that ICT innovations not only help with product innovations but also business model innovations. Business models refer to the design or architecture of mechanisms of value creation, delivery and capture employed by a firm (Teece 2010, Desyllas and Sako 2013). However, such innovation is difficult to measure since new business models are unlikely to qualify for a patent. Therefore, management scholars (Teece 2010, Amit and Zott 2012, Chan et al. 2018) define business model innovation as changes to individual components of a firm's business model (rather than one that focuses solely on changes to the overall architecture of a business model). In turn, they posit that business method patents are a clear measure of business model innovation

since they disclose activities that could be characterized as business model innovation or at the very least, underlying components of business model innovation.

Model IV of Table 5 presents the results for the regression of business model innovation (measured as the log of number of business method patents) on closeness to the ICT industry. Following Chan et al. (2018)), we identify patents belonging to IPC Class 705 (Data Processing – Financial, Business Practice, Management, or Cost/Price determination) as business method patents. These inventions, whose primary novel contribution is in creating or improving business operations⁷, fall into two categories: those related to customer service (including and not limited to those in the area of incentive programs, coupons, electronic shopping, cost/price determination, and reservations) and those related to operations/administration (such as operations research, health care management, inventory management, accounting, and shipping) (USPTO, 2017). We find that the coefficient of ICT-Closeness is positive and significant ($p < 0.01$), emphasizing that technology transfers from the ICT industry relate to greater business model innovations in the using industries.

Insert Table 5 here

The above set of results, together with the results for the impact of closeness to the ICT industry on innovation efficiency and recombinant capabilities, lend credence to the view of ICT as a general-purpose technology that is characterized by pervasive use as well as the ability to boost complementary innovations in a wide range of economic sectors (Bresnahan and Trajtenberg 1995, Lipsey et al. 2005).

7. Robustness Checks

We examine the robustness of our results to the endogenous nature of network formation and alternate industry classification. The details of these two analyses are provided in Online Appendix in Sections S1 and S2 respectively. The results are consistent with our main findings.

8. Conclusion

Our analyses lead to two important observations about innovation in modern industries. First, we find a significant increase over time in knowledge flows from the ICT industry across an array of industries

⁷ While business method patents include software patents, not all of them are software patents.

and fields. Automobiles, aerospace and defense, and pharmaceuticals, are but some of the many industries that extensively draw on the technologies and innovations of the ICT industry. Underlying the increase in technology flows from the ICT industry are significant increase in: (a) proportion of ICT patents, (b) complementarity among innovations, and (c) business model innovations in the consuming industry. Anecdotal evidence of this digitally biased shift in the content of innovation abounds in the form of electric vehicles and driverless cars in the automobile industry, jet engines that produce terabytes of data per flight to facilitate pre-emptive maintenance and repair, and the emergence of bioinformatics and computational biology as key tools in drug development. Yet, large-scale empirical evidence of the magnitude and nature of knowledge flows from the ICT industry is lacking. Our study adopts a network perspective to document and demonstrate both direct and indirect impacts of innovations in the ICT industry across a range of industrial sectors.

Given the characteristics of ICT technologies and the complementarities they share with other technologies, we find that closeness of a non-ICT industry to the ICT industries in the innovation network space positively relates to its innovation outcomes. Specifically, a unit standard deviation increase in closeness to the ICT industry yields a 10.2% improvement in innovative efficiency, 48.39% increase in recombinant reuse, and 9.09% increase in recombinant creation. The same increase in the closeness increases the cash flow spread by 5.56% and cash flow volatility by 5% for the industry.

It is valuable to ask why if the move towards incorporation of ICT technologies is associated with superior performance in all of these industries, what prevents certain industries from moving in that direction? Tushman and Anderson (1986)) point out that radical technological shifts might be competence destroying in nature and incumbents need to acquire new knowledge and capabilities that are consistent with this shift (Tushman and Anderson 1986, Gatignon et al. 2002, Lavie 2006). However, extant resources, processes and values of the firm engender inertia and resistance to change with core competencies acting as core rigidities during such shifts (Leonard - Barton 1992). Resource allocation processes in firms too may be biased towards addressing the needs of existing stakeholders (McElheran (2015); Mani et al. (2018)). The advancement of this work through a greater understanding of the micro-mechanisms that enable and constrain digital innovation represents an important direction

for future research. It would also be invaluable to explore shifts to products enabled through more ICT-centric R&D such as greater personalization engagement, and data-driven product design.

A paradigm shift to ICT-intensive R&D and the related increase in the centrality of the ICT industry has important implications for the future of competition in non-ICT industries. Specifically, IT companies are likely to transcend the boundaries of the ICT industry to emerge as formidable competitors that displace incumbent industry leaders in diverse industries. For example, in the automobile industry, technology companies like Uber, Google and Tesla are competing with traditional car manufacturers on multiple fronts and compelling the latter to reinvent their operating and business models. Similarly, in the pharmaceutical industry that engages in ICT-intensive R&D, technology companies such as Apple, IBM and Qualcomm Technologies are emerging as significant competition. These companies are exploiting shifts in patient engagement to extend sources of patient data beyond medical records to include apps, wearable devices and online communities that yield critical analytical insights for clinical decision support. Technology companies, with their superior technological absorptive capacity and agile operating models, may be better equipped to compete in this new business environment across diverse industries. Future research could examine such shifts in competition in industries that engage in ICT-centric R&D and allied shifts in sources of competitive advantage. Such research provides improved understanding of systematic differences in competitive dynamics, market valuations and productivity between industries that drive ICT-centric innovations and others.

Our study is subject to certain limitations. First, our data are limited in time since they only include patents granted through 2010. We expect that the technological trends of mobility and connectivity observed across industries reflect an amplification of the shifts in innovation documented in this study. Empirical analyses of patents granted after 2010 will help us demonstrate the continuity, if not strengthening, of the centrality of the ICT industry in modern R&D in recent years.

Second, our data are also limited to public firms in the United States (U.S.). Prior research (Kerr and Lincoln 2010, Arora et al. 2013) has emphasized a sizeable human resource advantage of the U.S. in the technology domain, and therefore, it is unclear whether the ICT-centricity of R&D documented in this study is idiosyncratic to the U.S. Future research could extend the current analyses to other countries to determine if the findings of our study extend to other countries.

Finally, our sample is restricted to patentable innovations alone. However, as Gilriches (1990) summarized, “*not all inventions are patentable, not all inventions are patented, and the inventions that are patented differ greatly in...the magnitude of inventive output associated with them*” (p. 1669). ICT technologies may not be patented for one of three reasons. First, given the significant costs of obtaining and enforcing the patent, firms typically tradeoff intellectual property (IP) protection offered by patents with other options such as trademarks, industrial designs, trade secret protection, etc. Second, certain ICT technologies, albeit valuable, may not be eligible for patent protection because of the difficulty involved in estimating the inventiveness of the patent over the existing state of the art. Software patents are especially susceptible to these challenges. Finally, the widespread and free use of ICT innovations, exemplified through open source software, suggests that ICT technologies may exist and be used without any IP protection at all. Hence, we recognize that our patent data do not encompass all innovations. In turn, our estimates of centrality of the ICT industry and the impact of the latter on innovation outcomes in using industries may be biased downward.

Despite these limitations, our study makes an important contribution in documenting and demonstrating significant technology flows from the ICT industry and the shifts they have engendered in the R&D function across a wide spectrum of industries. These shifts are blurring industry boundaries and engendering fundamental changes in competitive strategy. These emerging patterns further suggest that any changes to legislation and policy that impact innovation in the ICT industry, including proposed reforms in immigration laws, will impact the competitiveness of not just the ICT industry but rather, a range of industries traditionally considered unrelated to the ICT industry.

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